Supplementary Material: Statistical Tables and Figures

From Instrumental to Integral AI Learning: A Quasi-Experimental Evaluation of a Connectivist Framework for Multidimensional AI Literacy Among University Students

This document contains the participant flow diagram (Fig. S1) and the complete statistical tables (S2–S14) referenced in the main text. All quantities were computed from the analytic dataset ($N = 150$; eight classroom sections) with the analysis code available in the study repository (Python 3: NumPy, SciPy, pandas, statsmodels; R 4.5: lme4, lmerTest, pbkrtest, lavaan, semTools, ARTool). Random seeds are fixed in the code for full reproducibility. This document is anonymized for peer review.

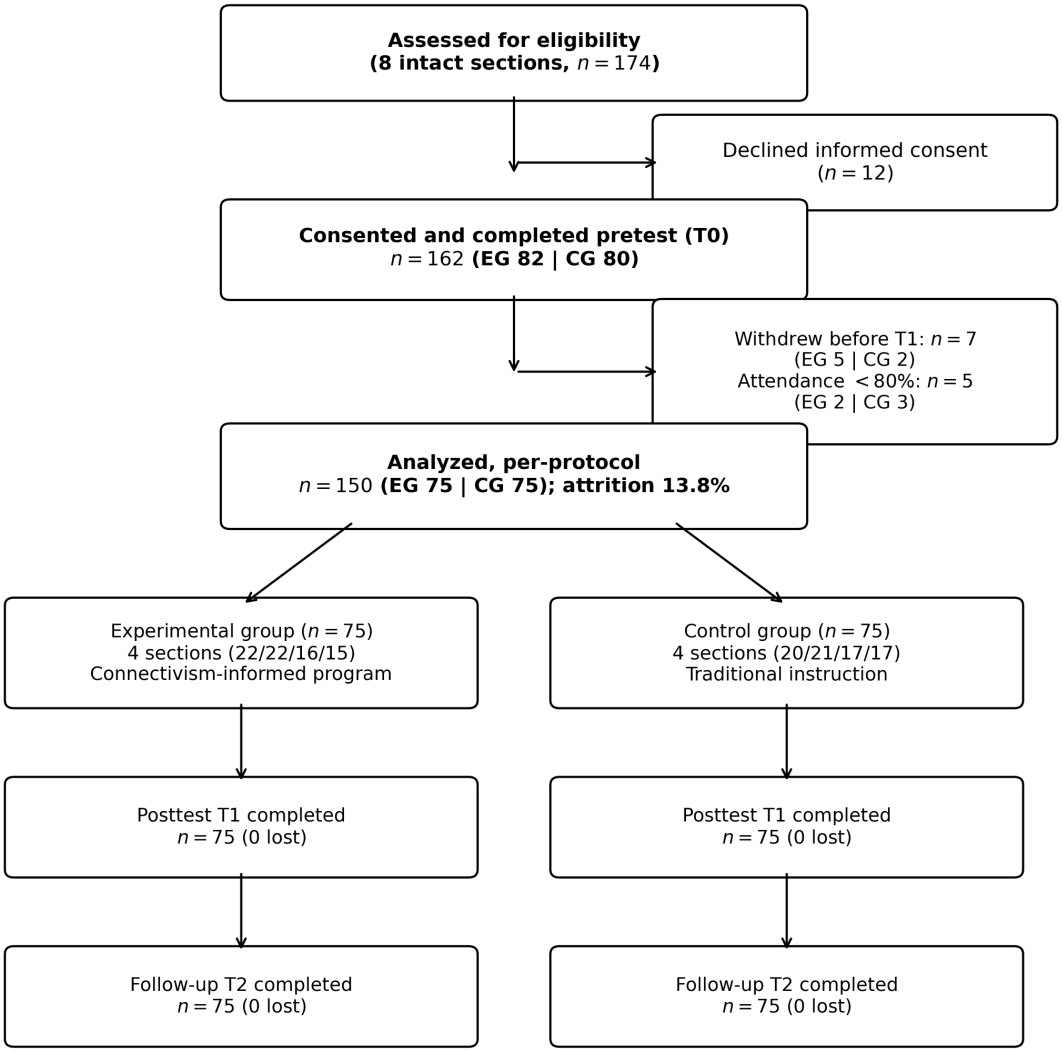

Figure S1: Participant flow diagram. EG = experimental group; CG = control group. Attrition did not differ by condition (Fisher's exact $p = .766$).

Table S2: Baseline comparison between completers ($n = 150$) and excluded participants with pretest data ($n = 12$).

Baseline variable	N compl.	M compl.	SD compl.	N excl.	M excl.	SD excl.	t (Welch)	p
Score_total_autopercepcion	150	2.521	0.729	12	2.566	0.812	-0.187	0.
Score_D1	150	2.58	0.886	12	2.709	1.0	-0.433	0.
Score_D2	150	2.013	0.825	12	2.057	0.984	-0.151	0.
Score_D3	150	2.928	0.886	12	2.863	0.895	0.242	0.
Score_D4	150	2.426	0.868	12	2.529	0.919	-0.376	0.
Score_conocimiento	150	8.7	4.218	12	7.917	4.144	0.629	0.
Score_conocimiento_D1	150	2.207	1.538	12	1.833	1.403	0.88	0.
Score_conocimiento_D2	150	1.247	1.231	12	1.417	1.084	-0.517	0.
Score_conocimiento_D3	150	3.427	1.873	12	3.0	1.651	0.852	0.
Score_conocimiento_D4	150	1.82	1.438	12	1.667	1.371	0.372	0.
Anio_nacimiento	150	2001.92	2.215	12	2001.917	2.353	0.005	0.
Ciclo_academico	150	5.28	2.168	12	5.25	2.417	0.042	0.
Atricion diferencial por grupo (Fisher exacto)	75			7			0.714	0.

Note. No comparison was significant (all $p \geq .39$). Differential attrition: 7/82 (EG) vs. 5/80 (CG), Fisher's exact $p = .766$, consistent with data missing completely at random.

Table S3: Section-level intraclass correlations by outcome and wave (one-way ANOVA estimator; exact F-based confidence intervals).

Outcome	Wave	ICC (section)	95		
Score_total_autopercepcion	T0	-0.0163	-0.0392	0.0937	1.0
Score_total_autopercepcion	T1	0.0478	-0.01	0.2759	1.846
Score_total_autopercepcion	T2	0.0277	-0.0194	0.2248	1.49
Score_conocimiento	T0	-0.0177	-0.0398	0.0892	1.0
Score_conocimiento	T1	0.2109	0.0753	0.5634	4.732
Score_conocimiento	T2	0.1256	0.0286	0.4358	3.222
Score_D1	T0	-0.0091	-0.0361	0.1172	1.0
Score_D1	T1	0.0466	-0.0105	0.2731	1.826
Score_D1	T2	0.0424	-0.0125	0.2625	1.75
Score_D2	T0	-0.0217	-0.0416	0.0755	1.0
Score_D2	T1	0.019	-0.0234	0.2011	1.336
Score_D2	T2	-0.0043	-0.0339	0.1324	1.0
Score_D3	T0	-0.0306	-0.0454	0.0443	1.0
Score_D3	T1	0.0089	-0.028	0.1724	1.158
Score_D3	T2	-0.0009	-0.0324	0.1429	1.0
Score_D4	T0	-0.0109	-0.0368	0.1116	1.0
Score_D4	T1	0.0363	-0.0154	0.2473	1.643
Score_D4	T2	0.0187	-0.0235	0.2002	1.33
Score_conocimiento_D1	T0	-0.0101	-0.0365	0.1141	1.0
Score_conocimiento_D1	T1	0.1242	0.0279	0.4334	3.199
Score_conocimiento_D1	T2	0.0996	0.0153	0.388	2.763
Score_conocimiento_D2	T0	-0.0461	-0.0521	-0.0143	1.0
Score_conocimiento_D2	T1	0.1336	0.0327	0.4495	3.364
Score_conocimiento_D2	T2	0.0298	-0.0184	0.2304	1.528
Score_conocimiento_D3	T0	-0.001	-0.0324	0.1426	1.0
Score_conocimiento_D3	T1	0.1279	0.0298	0.4398	3.264
Score_conocimiento_D3	T2	0.0767	0.0039	0.3415	2.357
Score_conocimiento_D4	T0	-0.0315	-0.0459	0.0408	1.0
Score_conocimiento_D4	T1	0.1173	0.0243	0.4212	3.077
Score_conocimiento_D4	T2	0.0685	-0.0	0.3239	2.213

Note. Design effect = $1 + (\bar{m} - 1)\,\text{ICC}$ with $\bar{m} \approx 18.75$; negative ICC estimates set to zero for the design effect.

Table S4: Standardized factor loadings of the Self-Perception Scale (WLSMV on polychoric correlations, pretest, $N = 150$).

Factor	Item	Std. loading	SE	p
D1	D1_01	0.8599	0.0273	<.001
D1	D1_02	0.8862	0.0292	<.001
D1	D1_03	0.8189	0.0382	<.001
D1	D1_04	0.83	0.0298	<.001
D1	D1_05	0.7794	0.0388	<.001
D1	D1_06	0.7032	0.0493	<.001
D2	D2_01	0.8962	0.0227	<.001
D2	D2_02	0.8434	0.0318	<.001
D2	D2_03	0.9018	0.0217	<.001
D2	D2_04	0.8616	0.0257	<.001
D2	D2_05	0.8383	0.0331	<.001
D2	D2_06	0.8069	0.0358	<.001
D3	D3_01	0.8333	0.0279	<.001
D3	D3_02	0.8807	0.0231	<.001
D3	D3_03	0.7251	0.0396	<.001
D3	D3_04	0.8494	0.0282	<.001
D3	D3_05	0.8276	0.0292	<.001
D3	D3_06	0.7059	0.0424	<.001
D3	D3_07	0.7409	0.0427	<.001
D3	D3_08	0.7587	0.0427	<.001
D4	D4_01	0.8208	0.0368	<.001
D4	D4_02	0.8322	0.0315	<.001
D4	D4_03	0.8581	0.0289	<.001
D4	D4_04	0.7447	0.0377	<.001
D4	D4_05	0.7715	0.0347	<.001
D4	D4_06	0.8195	0.0342	<.001

Note. Robust fit: CFI = .986, TLI = .984, RMSEA = .049 [.036, .061], SRMR = .053. No negative residual variances.

Table S4b: Factor correlations (WLSMV CFA).

Factor 1	Factor 2	r	SE	p
D1	D2	0.7657	0.0339	<.001
D1	D3	0.5491	0.0509	<.001
D1	D4	0.6459	0.0474	<.001
D2	D3	0.6094	0.0505	<.001
D2	D4	0.8387	0.0252	<.001
D3	D4	0.7428	0.0386	<.001

Table S4c: Average variance extracted (AVE) and composite reliability (CR) from standardized loadings (Fornell–Larcker).

Factor	k items	AVE	CR
D1	6	0.664	0.922
D2	6	0.737	0.944
D3	8	0.628	0.931
D4	6	0.654	0.919

Table S5: Longitudinal measurement invariance per dimension (Wu–Estabrook sequence for ordinal indicators; WLSMV; residuals of the same item correlated across waves).

Dimension	Model	χ^2 (scaled)	df	CFI	RMSEA	ΔCFI	Criterion met
D1	configural	115.9	114	1.0	0.0107		
D1	umbrales	142.1	138	0.9999	0.014	0.0	SI
D1	umbrales_cargas	150.0	148	1.0	0.0096	0.0	SI
D2	configural	161.2	114	0.9989	0.0527		
D2	umbrales	184.5	138	0.9989	0.0476	0.0	SI
D2	umbrales_cargas	195.1	148	0.9989	0.0462	0.0	SI
D3	configural	281.0	225	0.999	0.0409		
D3	umbrales	315.2	257	0.999	0.039	0.0	SI
D3	umbrales_cargas	330.1	271	0.9989	0.0383	0.0	SI
D4	configural	169.4	114	0.9994	0.0571		
D4	umbrales	201.4	138	0.9993	0.0555	-0.0001	SI
D4	umbrales_cargas	213.3	148	0.9993	0.0544	0.0	SI

Note. Models: configural; equal thresholds; equal thresholds and loadings. Criterion: $|\Delta\text{CFI}| \leq .01$.

Table S6: Mixed 2×3 ANOVAs with Greenhouse–Geisser corrected p values and η_p^2 with 90% confidence intervals.

Outcome	Effect	F	df1	df2	p (GG)	ε applied	η_p^2	90	
Autopercepcion total	Grupo	3.19	1	148	0.076	1.0	0.021	0.0	0.073
Autopercepcion total	Tiempo	777.0	2	296	<.0001	0.846	0.84	0.815	0.858
Autopercepcion total	Grupo x Tiempo	239.91	2	296	<.0001	0.846	0.618	0.564	0.66
Autopercepcion D1 (conceptual)	Grupo	1.76	1	148	0.1867	1.0	0.012	0.0	0.056
Autopercepcion D1 (conceptual)	Tiempo	247.48	2	296	<.0001	0.855	0.626	0.572	0.666
Autopercepcion D1 (conceptual)	Grupo x Tiempo	80.01	2	296	<.0001	0.855	0.351	0.279	0.411
Autopercepcion D2 (tecnica)	Grupo	1.76	1	148	0.1861	1.0	0.012	0.0	0.056
Autopercepcion D2 (tecnica)	Tiempo	137.3	2	296	<.0001	0.872	0.481	0.414	0.534
Autopercepcion D2 (tecnica)	Grupo x Tiempo	30.98	2	296	<.0001	0.872	0.173	0.109	0.233
Autopercepcion D3 (etica)	Grupo	2.36	1	148	0.1263	1.0	0.016	0.0	0.063
Autopercepcion D3 (etica)	Tiempo	248.76	2	296	<.0001	0.823	0.627	0.573	0.668
Autopercepcion D3 (etica)	Grupo x Tiempo	84.88	2	296	<.0001	0.823	0.364	0.292	0.424
Autopercepcion D4 (colaborativa)	Grupo	3.04	1	148	0.0831	1.0	0.02	0.0	0.071
Autopercepcion D4 (colaborativa)	Tiempo	201.01	2	296	<.0001	0.797	0.576	0.517	0.621
Autopercepcion D4 (colaborativa)	Grupo x Tiempo	68.94	2	296	<.0001	0.797	0.318	0.245	0.379
Conocimiento total	Grupo	14.77	1	148	0.0002	1.0	0.091	0.03	0.169
Conocimiento total	Tiempo	105.34	2	296	<.0001	0.964	0.416	0.345	0.473
Conocimiento total	Grupo x Tiempo	52.9	2	296	<.0001	0.964	0.263	0.192	0.326
Conocimiento D1	Grupo	8.85	1	148	0.0034	1.0	0.056	0.011	0.125
Conocimiento D1	Tiempo	27.6	2	296	<.0001	0.959	0.157	0.096	0.216
Conocimiento D1	Grupo x Tiempo	10.57	2	296	<.0001	0.959	0.067	0.025	0.113
Conocimiento D2	Grupo	8.33	1	148	0.0045	1.0	0.053	0.01	0.121
Conocimiento D2	Tiempo	23.71	2	296	<.0001	0.976	0.138	0.08	0.196
Conocimiento D2	Grupo x Tiempo	11.24	2	296	<.0001	0.976	0.071	0.028	0.118
Conocimiento D3	Grupo	12.5	1	148	0.0005	1.0	0.078	0.022	0.153
Conocimiento D3	Tiempo	30.87	2	296	<.0001	0.989	0.173	0.109	0.233
Conocimiento D3	Grupo x Tiempo	14.68	2	296	<.0001	0.989	0.09	0.042	0.141
Conocimiento D4	Grupo	6.96	1	148	0.0092	1.0	0.045	0.006	0.11
Conocimiento D4	Tiempo	26.32	2	296	<.0001	0.998	0.151	0.09	0.21
Conocimiento D4	Grupo x Tiempo	19.38	2	296	<.0001	0.998	0.116	0.061	0.171

Table S7: Assumption checks: Shapiro–Wilk (minimum p across waves, by group), Levene per occasion, Mauchly sphericity and epsilon corrections.

Outcome	Shapiro p_{min} EG	Shapiro p_{min} CG	Levene p T0	Levene p T1	Levene p T2	Mauchly W	Mau
Autopercepcion total	0.1586	0.1228	0.3719	0.7682	0.9272	0.8173	0.0
Autopercepcion D1 (conceptual)	0.0245	0.0233	0.805	0.1176	0.3935	0.8304	0.0
Autopercepcion D2 (tecnica)	0.0024	0.0001	0.4364	0.9363	0.9976	0.8529	0.0
Autopercepcion D3 (etica)	0.0019	0.195	0.9155	0.9009	0.8585	0.7849	0.0
Autopercepcion D4 (colaborativa)	0.061	0.0087	0.3583	0.6224	0.9831	0.7445	0.0
Conocimiento total	0.1152	0.0024	0.5611	0.176	0.6066	0.9631	0.062
Conocimiento D1	0.0	0.0003	0.5437	0.1082	0.0765	0.9574	0.040
Conocimiento D2	0.0	0.0	0.4089	0.1879	0.1005	0.9758	0.164

Table S8: Non-parametric sensitivity I: rank-transformed ANCOVA (Conover) for the group contrast at posttest, with conventional and cluster-robust inference.

Outcome	b (ranks)	t (classical)	p (classical)	SE CR2	df Satt.	p (CR2)
Score_total_autopercepcion	19.7	16.93	<.0001	0.94	5.87	<.0001
Score_conocimiento	36.07	9.2	<.0001	2.34	5.87	<.0001
Score_D1	18.66	10.75	<.0001	2.36	5.87	0.0002
Score_D2	12.55	6.48	<.0001	0.96	5.86	<.0001
Score_D3	18.66	10.27	<.0001	1.44	5.86	<.0001
Score_D4	17.39	10.38	<.0001	1.58	5.87	<.0001
Score_conocimiento_D1	24.83	4.38	<.0001	6.74	5.86	0.0107
Score_conocimiento_D2	28.32	4.65	<.0001	7.32	5.87	0.0086
Score_conocimiento_D3	30.41	5.75	<.0001	5.33	5.86	0.0014
Score_conocimiento_D4	30.54	5.7	<.0001	4.68	5.86	0.0007

Table S8b: Non-parametric sensitivity II: aligned-rank-transform (ART) mixed ANOVAs for the primary outcomes.

Outcome	Effect	F	df1	df2	p
Score_total_autopercepcion	Grupo	3.57	1	148	0.06087
Score_total_autopercepcion	Tiempo	525.09	2	296	4.403e-98
Score_total_autopercepcion	Grupo:Tiempo	204.47	2	296	1.678e-56
Score_conocimiento	Grupo	14.23	1	148	0.0002335
Score_conocimiento	Tiempo	98.28	2	296	1.85e-33
Score_conocimiento	Grupo:Tiempo	48.11	2	296	8.113e-19

Table S9: Attrition sensitivity bounds. The 12 excluded participants have pretest data only; posttest values were imputed under extreme counterfactual scenarios.

Outcome	b (completers)	SE	b (worst case)	p (worst case)	b (best case)	b (MAR-plausible)	Relative chang
Score_total_autopercepcion	0.39	0.02	0.32	<1e-6	0.398	0.389	-18.0
Score_conocimiento	4.987	0.519	4.179	<1e-6	5.039	4.987	-16.2

Note. Worst case: excluded EG participants gain nothing; excluded CG participants gain the EG mean. Best case: mirror assumptions. MAR-plausible: each excluded case receives its own arm's mean gain.

Table S10: Correlations between self-perception and objective knowledge scores (full sample, $N = 150$).

Wave	Dimension	Pearson r	p	Spearman ρ	p
T0	D1	0.755	<.001	0.766	<.001
T0	D2	0.739	<.001	0.629	<.001
T0	D3	0.758	<.001	0.756	<.001
T0	D4	0.76	<.001	0.743	<.001
T0	Total	0.87	<.001	0.864	<.001
T1	D1	0.819	<.001	0.818	<.001
T1	D2	0.756	<.001	0.739	<.001
T1	D3	0.807	<.001	0.818	<.001
T1	D4	0.838	<.001	0.84	<.001
T1	Total	0.907	<.001	0.91	<.001
T2	D1	0.809	<.001	0.814	<.001
T2	D2	0.761	<.001	0.681	<.001
T2	D3	0.825	<.001	0.817	<.001
T2	D4	0.809	<.001	0.805	<.001
T2	Total	0.911	<.001	0.907	<.001

Table S11: Omnibus Group × Moderator interaction tests on the primary outcomes (ANCOVA framework, exploratory).

Outcome	Moderator	n	F interaction	df1	df2	p
Score_total_autopercepcion	Facultad	150	1.024	5	137	0.4059
Score_total_autopercepcion	Genero (3 categorias)	150	0.771	2	143	0.4646
Score_total_autopercepcion	Genero (binario, sens.)	143	1.173	1	138	0.2807
Score_total_autopercepcion	Experiencia previa (4 cat.)	150	0.71	3	141	0.5477
Score_total_autopercepcion	Experiencia (colapsada, sens.)	150	1.075	2	143	0.3442
Score_conocimiento	Facultad	150	1.209	5	137	0.3081
Score_conocimiento	Genero (3 categorias)	150	0.119	2	143	0.8877
Score_conocimiento	Genero (binario, sens.)	143	0.12	1	138	0.7298
Score_conocimiento	Experiencia previa (4 cat.)	150	0.628	3	141	0.5981
Score_conocimiento	Experiencia (colapsada, sens.)	150	0.228	2	143	0.7961

Note. Sensitivity codings: binary gender excludes non-binary participants ($n = 7$; one in the CG); collapsed experience merges Intermediate and Advanced.

Table S11b: Subgroup estimates at posttest (hypothesis-generating; no omnibus interaction was significant).

Moderator	Category	Scale	n_{EG}	n_{CG}	M_{EG}	M_{CG}	d
Facultad	Arte y Humanidades	Autopercepcion	6	9	3.49	2.46	1.312
Facultad	Arte y Humanidades	Conocimiento	6	9	16.33	7.89	1.839
Facultad	Ciencias	Autopercepcion	17	13	2.76	3.0	-0.347
Facultad	Ciencias	Conocimiento	17	13	12.41	11.69	0.143
Facultad	Ciencias Sociales	Autopercepcion	9	11	2.91	2.66	0.253
Facultad	Ciencias Sociales	Conocimiento	9	11	14.78	9.18	0.923
Facultad	Ciencias de la Salud	Autopercepcion	10	8	3.1	2.29	1.125
Facultad	Ciencias de la Salud	Conocimiento	10	8	15.2	6.88	1.465
Facultad	Educación	Autopercepcion	16	14	3.04	2.69	0.44
Facultad	Educación	Conocimiento	16	14	13.69	9.14	0.842
Facultad	Ingeniería	Autopercepcion	17	20	3.22	2.68	0.774
Facultad	Ingeniería	Conocimiento	17	20	16.29	10.3	1.262
Genero	Femenino	Autopercepcion	38	35	3.0	2.7	0.362
Genero	Femenino	Conocimiento	38	35	14.26	9.31	0.869
Genero	Masculino	Autopercepcion	31	39	3.15	2.63	0.737

Moderator	Category	Scale	n_{EG}	n_{CG}	M_{EG}	M_{CG}	d
Genero	Masculino	Conocimiento	31	39	14.84	9.69	1.048
Genero	No binario	Autopercepcion	6	1	2.73	2.88	
Genero	No binario	Conocimiento	6	1	14.67	9.0	
Experiencia_previa_IA	Avanzada	Autopercepcion	6	7	3.11	2.74	0.441
Experiencia_previa_IA	Avanzada	Conocimiento	6	7	13.83	8.57	1.065
Experiencia_previa_IA	Básica	Autopercepcion	48	32	3.12	2.8	0.415
Experiencia_previa_IA	Básica	Conocimiento	48	32	15.04	10.09	0.929
Experiencia_previa_IA	Intermedia	Autopercepcion	7	20	2.71	2.66	0.067
Experiencia_previa_IA	Intermedia	Conocimiento	7	20	14.57	10.1	0.916
Experiencia_previa_IA	Ninguna	Autopercepcion	14	16	2.93	2.36	0.641
Experiencia_previa_IA	Ninguna	Conocimiento	14	16	13.07	8.0	0.889

Table S12: Exploratory structural equation model: fit indices and ratio of sample size to free parameters (WLSMV).

Index	Value
npar	150.0
chisq.scaled	466.7298
df.scaled	364.0
cfi.scaled	0.9716
tli.scaled	0.9705
rmsea.scaled	0.0435
rmsea.ci.lower.scaled	0.0307
rmsea.ci.upper.scaled	0.0547
srmr	0.0505
N	150.0
N_q	1.0

Note. $N{:}q = 1.0{:}1$ including thresholds ($\approx 3.3{:}1$ excluding them); reported as a descriptive complement only.

Table S12b: Exploratory SEM: standardized structural paths with 95% confidence intervals.

Endogenous	Predictor	β (std)	SE	p	95	
Improvement	D1	0.1134	0.0818	0.1657	-0.047	0.2737
Improvement	D2	0.0355	0.1224	0.7721	-0.2045	0.2754
Improvement	D3	0.0637	0.0666	0.3387	-0.0668	0.1943
Improvement	D4	0.0404	0.1074	0.7069	-0.1702	0.2509
Improvement	Group	0.8449	0.0183	0.0	0.809	0.8807
Knowledge	D1	0.354	0.0589	0.0	0.2385	0.4695
Knowledge	D2	0.1655	0.1046	0.1137	-0.0395	0.3705
Knowledge	D3	0.2341	0.0579	0.0001	0.1205	0.3476
Knowledge	D4	0.3113	0.1017	0.0022	0.112	0.5106

Table S13: Descriptive statistics by wave, group, instrument, and dimension.

Wave	Group	Scale	Dim	N	Mean	SD	Median	Min	Max	Skew.	Kurt.
T0	Experimental	Autopercepcion	D1	75	2.545	0.897	2.5	1.0	4.33	0.103	-0.823
T0	Experimental	Autopercepcion	D2	75	2.038	0.786	2.0	1.0	4.67	0.633	0.385
T0	Experimental	Autopercepcion	D3	75	2.915	0.888	2.88	1.0	4.62	-0.298	-0.497
T0	Experimental	Autopercepcion	D4	75	2.427	0.835	2.5	1.0	4.67	0.227	-0.427
T0	Experimental	Autopercepcion	Total	75	2.514	0.705	2.62	1.0	4.12	-0.234	-0.529
T0	Experimental	Conocimiento	D1	75	2.227	1.512	2.0	0.0	6.0	0.532	-0.583
T0	Experimental	Conocimiento	D2	75	1.267	1.212	1.0	0.0	6.0	1.358	2.55
T0	Experimental	Conocimiento	D3	75	3.493	1.727	3.0	0.0	7.0	0.194	-0.661
T0	Experimental	Conocimiento	D4	75	1.733	1.359	2.0	0.0	5.0	0.459	-0.423
T0	Experimental	Conocimiento	Total	75	8.72	4.099	9.0	1.0	19.0	0.206	-0.476
T0	Control	Autopercepcion	D1	75	2.616	0.88	2.67	1.0	4.17	-0.146	-0.932
T0	Control	Autopercepcion	D2	75	1.987	0.867	2.0	1.0	4.5	0.822	0.068
T0	Control	Autopercepcion	D3	75	2.941	0.89	3.0	1.12	4.88	0.194	-0.401
T0	Control	Autopercepcion	D4	75	2.425	0.906	2.33	1.0	4.17	0.225	-1.023
T0	Control	Autopercepcion	Total	75	2.527	0.757	2.54	1.08	4.12	0.194	-0.914
T0	Control	Conocimiento	D1	75	2.187	1.574	2.0	0.0	6.0	0.233	-0.588
T0	Control	Conocimiento	D2	75	1.227	1.258	1.0	0.0	5.0	1.0	0.606
T0	Control	Conocimiento	D3	75	3.36	2.018	3.0	0.0	8.0	0.325	-0.801
T0	Control	Conocimiento	D4	75	1.907	1.517	2.0	0.0	5.0	0.533	-0.65
T0	Control	Conocimiento	Total	75	8.68	4.362	8.0	2.0	18.0	0.329	-0.791
T1	Experimental	Autopercepcion	D1	75	3.135	1.041	3.17	1.17	5.0	-0.093	-0.969
T1	Experimental	Autopercepcion	D2	75	2.44	0.892	2.33	1.0	5.0	0.549	-0.08
T1	Experimental	Autopercepcion	D3	75	3.477	0.962	3.62	1.25	4.88	-0.646	-0.318
T1	Experimental	Autopercepcion	D4	75	2.975	0.97	3.0	1.17	5.0	-0.097	-0.922
T1	Experimental	Autopercepcion	Total	75	3.043	0.793	3.15	1.19	4.58	-0.306	-0.61
T1	Experimental	Conocimiento	D1	75	3.587	1.967	4.0	0.0	6.0	-0.323	-1.084
T1	Experimental	Conocimiento	D2	75	2.573	1.637	2.0	0.0	6.0	0.185	-0.806
T1	Experimental	Conocimiento	D3	75	5.227	2.172	6.0	0.0	8.0	-0.811	-0.005
T1	Experimental	Conocimiento	D4	75	3.147	1.843	3.0	0.0	6.0	-0.047	-1.0
T1	Experimental	Conocimiento	Total	75	14.533	5.639	15.0	2.0	25.0	-0.182	-0.859
T1	Control	Autopercepcion	D1	75	2.773	0.87	2.83	1.0	4.5	-0.245	-0.938
T1	Control	Autopercepcion	D2	75	2.114	0.906	2.0	1.0	4.67	0.824	-0.084
T1	Control	Autopercepcion	D3	75	3.086	0.918	3.12	1.25	4.88	0.021	-0.737
T1	Control	Autopercepcion	D4	75	2.553	0.929	2.5	1.0	4.5	0.2	-1.058
T1	Control	Autopercepcion	Total	75	2.666	0.752	2.58	1.19	4.23	0.158	-0.94
T1	Control	Conocimiento	D1	75	2.48	1.605	2.0	0.0	6.0	0.096	-1.061
T1	Control	Conocimiento	D2	75	1.453	1.445	1.0	0.0	5.0	0.742	-0.548
T1	Control	Conocimiento	D3	75	3.653	1.899	3.0	0.0	8.0	0.399	-0.613
T1	Control	Conocimiento	D4	75	1.92	1.558	2.0	0.0	6.0	0.587	-0.325
T1	Control	Conocimiento	Total	75	9.507	4.847	8.0	2.0	20.0	0.398	-0.95
T2	Experimental	Autopercepcion	D1	75	3.078	1.004	3.17	1.17	4.83	-0.147	-0.902
T2	Experimental	Autopercepcion	D2	75	2.338	0.854	2.17	1.0	4.5	0.468	-0.458
T2	Experimental	Autopercepcion	D3	75	3.387	0.934	3.5	1.12	4.88	-0.587	-0.416
T2	Experimental	Autopercepcion	D4	75	2.926	0.937	3.0	1.33	5.0	0.015	-0.858
T2	Experimental	Autopercepcion	Total	75	2.967	0.766	3.08	1.27	4.46	-0.293	-0.584
T2	Experimental	Conocimiento	D1	75	3.4	1.724	4.0	0.0	6.0	-0.395	-0.798
T2	Experimental	Conocimiento	D2	75	2.053	1.61	2.0	0.0	6.0	0.676	-0.283

Wave	Group	Scale	Dim	N	Mean	SD	Median	Min	Max	Skew.	Kurt.
T2	Experimental	Conocimiento	D3	75	4.88	1.823	5.0	0.0	8.0	-0.589	-0.421
T2	Experimental	Conocimiento	D4	75	2.907	1.733	3.0	0.0	6.0	0.003	-0.986
T2	Experimental	Conocimiento	Total	75	13.24	5.122	13.0	2.0	25.0	0.014	-0.336
T2	Control	Autopercepcion	D1	75	2.769	0.892	2.83	1.0	4.5	-0.165	-0.934
T2	Control	Autopercepcion	D2	75	2.158	0.891	2.0	1.0	4.83	0.853	0.139
T2	Control	Autopercepcion	D3	75	3.07	0.892	3.12	1.38	4.88	0.086	-0.896
T2	Control	Autopercepcion	D4	75	2.576	0.91	2.5	1.0	4.5	0.158	-1.054
T2	Control	Autopercepcion	Total	75	2.676	0.733	2.62	1.23	4.27	0.148	-0.921
T2	Control	Conocimiento	D1	75	2.507	1.446	2.0	0.0	6.0	0.446	-0.576
T2	Control	Conocimiento	D2	75	1.493	1.537	1.0	0.0	6.0	1.475	1.65
T2	Control	Conocimiento	D3	75	3.667	2.022	4.0	0.0	8.0	0.067	-0.666
T2	Control	Conocimiento	D4	75	2.173	1.545	2.0	0.0	6.0	0.371	-0.567
T2	Control	Conocimiento	Total	75	9.84	4.728	9.0	1.0	22.0	0.326	-0.56

Table S14: Item analysis of the Objective Knowledge Test: difficulty per wave and corrected item–total correlation at pretest.

Item	Dim	p_{T0}	p_{T1}	p_{T2}	r_{it} (T0)
OBJ_D1_01	D1	0.587	0.7	0.66	0.278
OBJ_D1_02	D1	0.513	0.667	0.733	0.246
OBJ_D1_03	D1	0.427	0.533	0.56	0.261
OBJ_D1_04	D1	0.32	0.46	0.48	0.198
OBJ_D1_05	D1	0.213	0.353	0.273	0.348
OBJ_D1_06	D1	0.147	0.32	0.247	0.117
OBJ_D2_01	D2	0.373	0.547	0.573	0.303
OBJ_D2_02	D2	0.333	0.507	0.347	0.451
OBJ_D2_03	D2	0.22	0.4	0.313	0.253
OBJ_D2_04	D2	0.14	0.267	0.233	0.131
OBJ_D2_05	D2	0.087	0.173	0.153	0.23
OBJ_D2_06	D2	0.093	0.12	0.153	0.191
OBJ_D3_01	D3	0.7	0.873	0.78	0.247
OBJ_D3_02	D3	0.627	0.793	0.713	0.148
OBJ_D3_03	D3	0.52	0.607	0.633	0.339
OBJ_D3_04	D3	0.493	0.58	0.673	0.304
OBJ_D3_05	D3	0.34	0.467	0.473	0.309
OBJ_D3_06	D3	0.273	0.44	0.393	0.336
OBJ_D3_07	D3	0.26	0.34	0.327	0.323
OBJ_D3_08	D3	0.213	0.34	0.28	0.156
OBJ_D4_01	D4	0.593	0.647	0.713	0.4
OBJ_D4_02	D4	0.413	0.493	0.587	0.413
OBJ_D4_03	D4	0.293	0.473	0.5	0.308
OBJ_D4_04	D4	0.22	0.387	0.26	0.339
OBJ_D4_05	D4	0.16	0.287	0.233	0.249
OBJ_D4_06	D4	0.14	0.247	0.247	0.173